MECH321 Dynamics of Engineering Systems

Lecture 5
Frequency Response

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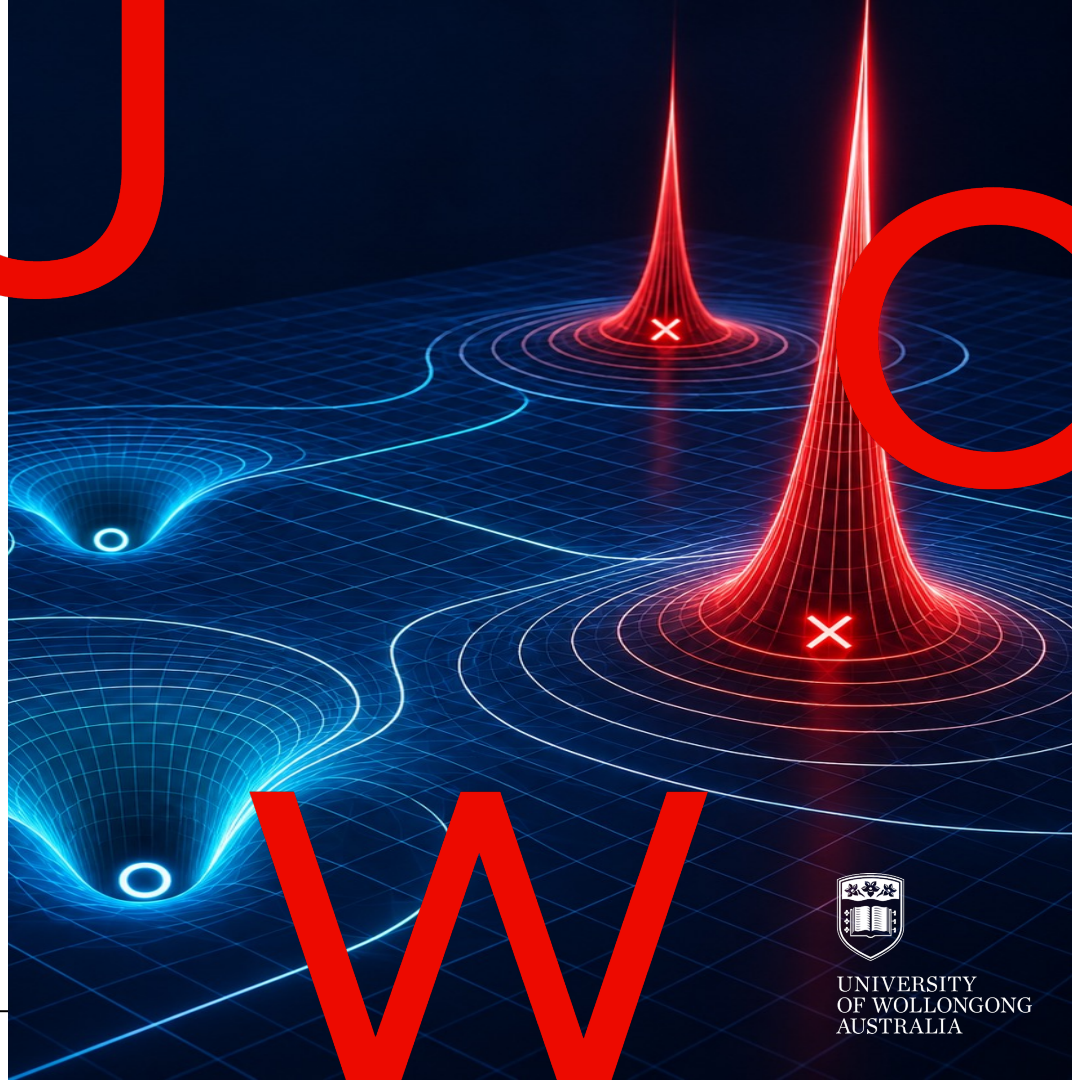


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PART 1

Frequency response and Bode plots

01 Amplitude and phase

02 Decibels and factors

03 Bode sketches

Sinusoidal input and steady-state output

EXAMPLE

For a stable LTI system, the output keeps the input frequency

$$u(t) = A\sin(\omega t)$$

$$y_{ss}(t) = A|G(j\omega)|\sin\left(\omega t + \phi(\omega)\right)$$

A: input amplitude

$|G(j\omega)|$: output/input amplitude ratio

$y_{ss}(t)$: long-time periodic output

ω: input angular frequency (rad/s)

$\phi(\omega) = \arg G(j\omega)$: output phase shift

Output frequency remains ω.

Why substitute s = jω?

$$\frac{d^n}{dt^n}e^{j\omega t} = (j\omega)^n e^{j\omega t}$$

$$Ue^{j\omega t} \longrightarrow G(j\omega)Ue^{j\omega t}$$

Every derivative multiplies by jω. Insert into the ODE and cancel the common exponential.

Worked: amplitude and phase

EXAMPLE

Question

$$G(s) = \frac{5}{1 + 0.2s}, \qquad u(t) = 2\sin(10t)$$

Find the steady-state output amplitude, phase shift and time-domain waveform.

1 Set $s = j\omega$ and substitute $\omega = 10$ rad/s

$$0.2(j10) = 2j, \qquad G(j10) = \frac{5}{1 + 2j}$$

Method 1 Quotient rule

$$|G(j10)| = 5/\sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.236$$

$$\phi = 0° - \arctan(2) = -63.4°$$

Method 2 Polar form

$$1 + 2j = \sqrt{5}\, e^{j \arctan(2)}$$

$$G(j10) = \sqrt{5}\, e^{-j \arctan(2)}$$

2 Apply that gain and phase to the input

$$y_{ss}(t) = 2\sqrt{5}\sin(10t - 63.4°) \approx 4.472\sin(10t - 63.4°)$$

Check The gain is below the DC value 5, and the phase is a lag. In polar form the angle is arctan(2), not 2 radians.

Decibels as amplitude ratios

$$M_{\mathrm{dB}}(\omega) = 20\log_{10}|G(j\omega)|$$

Amplitude ratio $|G| \leftrightarrow$ dB.

$\lvert G\rvert$	**0.1**	**0.5**	**1**	**2**	**10**
Magnitude	−20 dB	−6.02 dB	0 dB	+6.02 dB	+20 dB

0 dB means equal amplitudes. Negative dB means attenuation; positive dB means amplification.

The Bode sketching rule: products become sums

$$20\log_{10}|G_1G_2| = 20\log_{10}|G_1| + 20\log_{10}|G_2|$$

If $G = G_1G_2$, add the two dB curves point by point; add their phase curves too. Example: $2\times0.5 = 1$, so $+6.02 - 6.02 = 0$ dB.

First-order Bode curve

EXAMPLE

How the curve is made

$$G(j\omega) = \frac{1}{1 + 0.2j\omega}, \qquad \omega > 0$$

Sweep $\omega > 0$: $s = j\omega$.

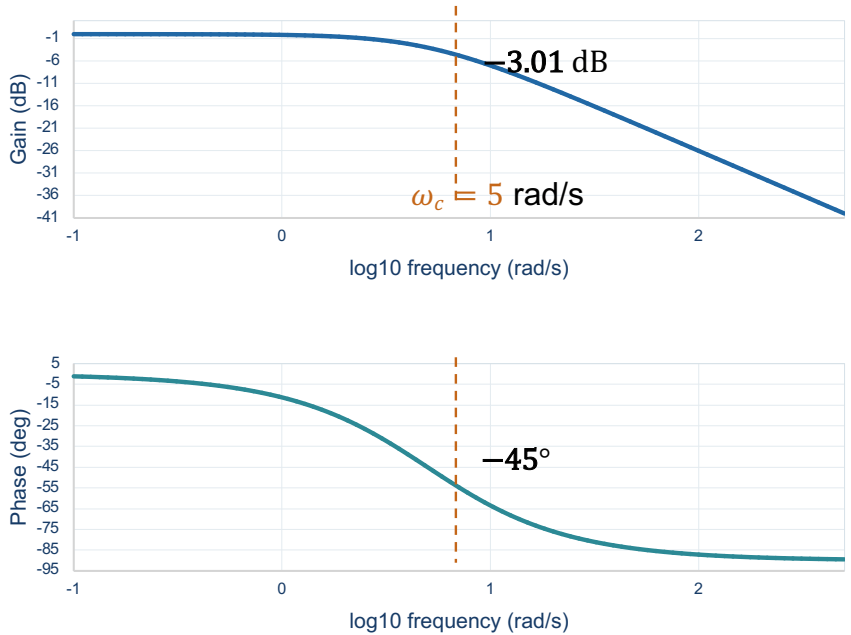
Top graph: $20\log_{10}|G(j\omega)|$ in dB.

Bottom graph: arg $G(\mathrm{j}\omega)$ in degrees.

Horizontal axis: logarithmic ω

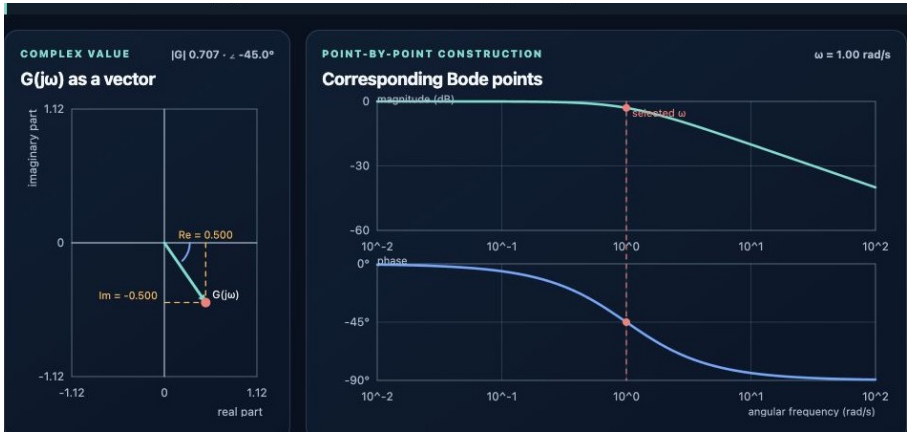
At $\omega_c = 5$ rad/s: −3.01 dB, −45°

Vertical axes use linear scales.



Live demo: one frequency, two Bode points

OPEN LIVE DEMO ↗



Course home: Demo 4.

SET UP Demo 1: $K = 1$, $\tau = 1$ s. Apply values. Open Demo 4: First order.

1 Try $\omega = 0.1$, 1 and 10 rad/s. Apply frequency.

2 Read the complex value, dB and phase. Find the highlighted Bode points.

MATLAB alternative (Control System Toolbox)

```
s=tf('s'); G=1/(s+1); w=1;
H=evalfr(G,1i*w);
disp([real(H), imag(H), ...
  20*log10(abs(H)), angle(H)*180/pi])
figure; bode(G); grid on
figure; compass(H); grid on
```

CHECK At $\omega = 1$: $G(j\omega) = 0.5 - j0.5$, $|G| = 0.707$ and phase=−45°.

Five Bode building blocks

Factor	Magnitude behaviour	Slope	Phase
gain $K\ (K > 0)$	Flat: $20\log_{10} K$ dB	**0**	0°
integrator $1/s$	Falls at every ω	**−20**	−90°
first-order pole $1/(1 + s/\omega_p)$	Flat, then falls	**−20**	0° to −90°
first-order zero $1 + s/\omega_z$	Flat, then rises	**+20**	0° to +90°
second-order poles $\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$	May peak; then falls	**−40**	0° to −180°

Slopes are in dB per decade, **after the corner unless stated**. Near a corner or resonance, use the exact factor.

Build a Bode plot factor by factor

1 Factor the transfer function

Separate gain, origin factors, zeros and poles.

2 Sketch each factor on the same ω axis

Set the initial magnitude and slope.

3 Mark every corner frequency

Zero: +20 dB/dec; pole: −20 dB/dec after its corner.

4 Combine the factors

Add factor magnitudes in dB and factor phases in degrees.

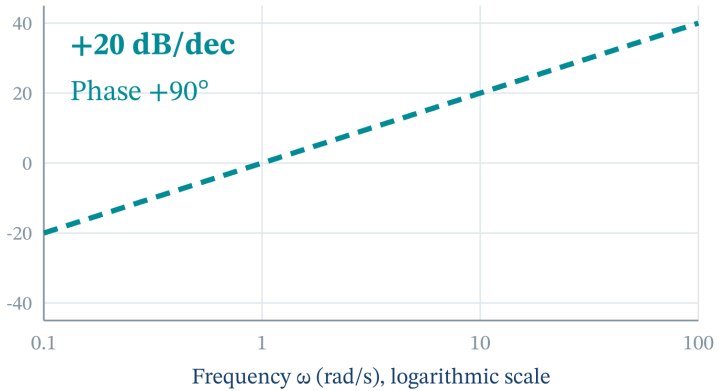
Bode building blocks: gain and origin factors

CORE

Constant gain K = 10 shifts magnitude by +20 dB at every frequency; phase is 0°.

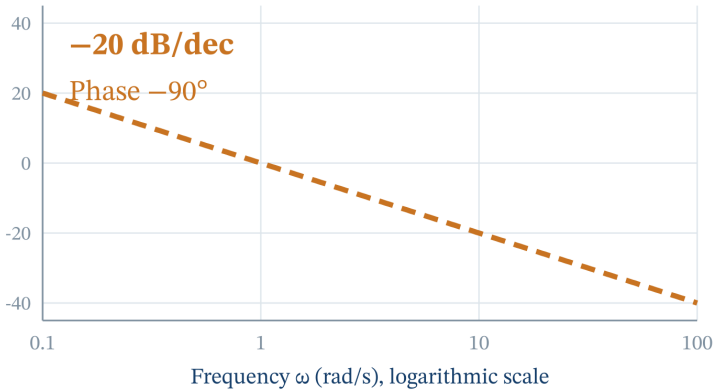
Differentiator

$H(s) = \frac{s}{\omega_0}$ $\omega_0 = 1$ rad/s



Integrator

$H(s) = \frac{\omega_0}{s}$ $\omega_0 = 1$ rad/s



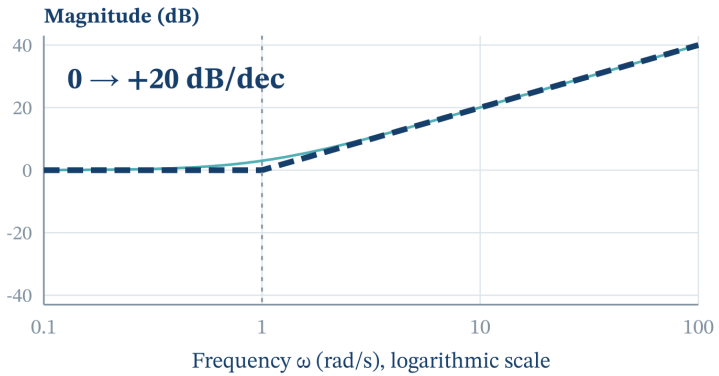
These two dashed lines are exact. The constant gain changes level but never changes slope.

First-order zero and pole

First-order zero

$$H_z(s) = \frac{s + \omega_z}{\omega_z}$$

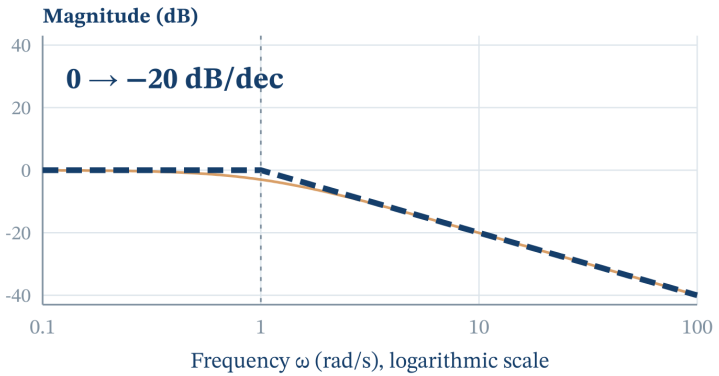
$\omega_z = 1$ rad/s



First-order pole

$$H_p(s) = \frac{\omega_p}{s + \omega_p}$$

$\omega_p = 1$ rad/s

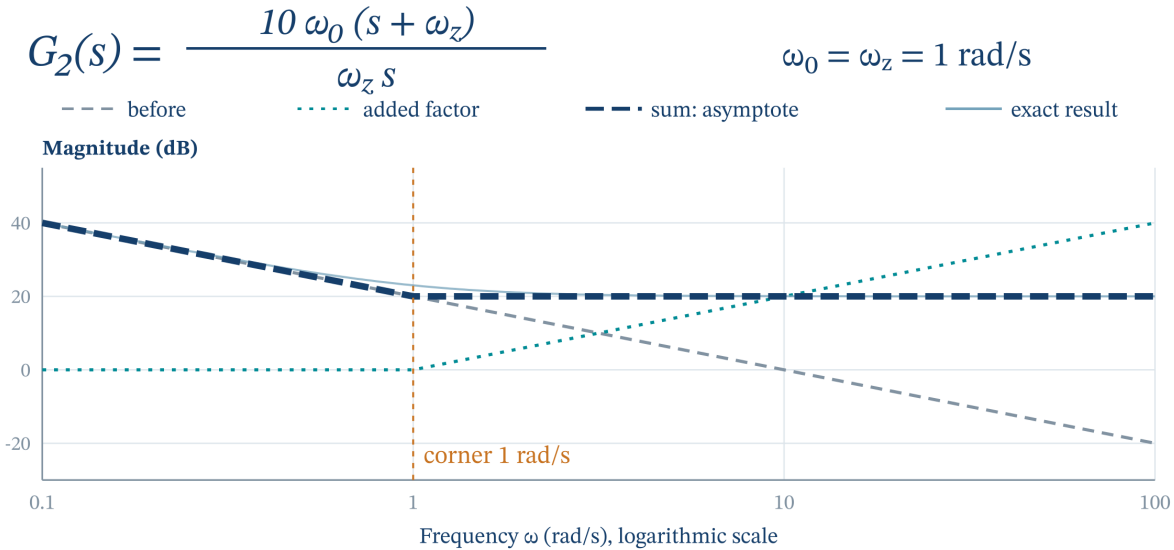


asymptote (used for addition) thin exact curve

ω_z: zero corner; ω_p: pole corner. Both are 1 rad/s here; exact magnitude at the corner is ±3.01 dB.

Add a zero at 1 rad/s

$$G_2(s) = \frac{10\,\omega_0\,(s+\omega_z)}{\omega_z\,s} \qquad \omega_0 = \omega_z = 1 \text{ rad/s}$$



ADD MAGNITUDES IN dB

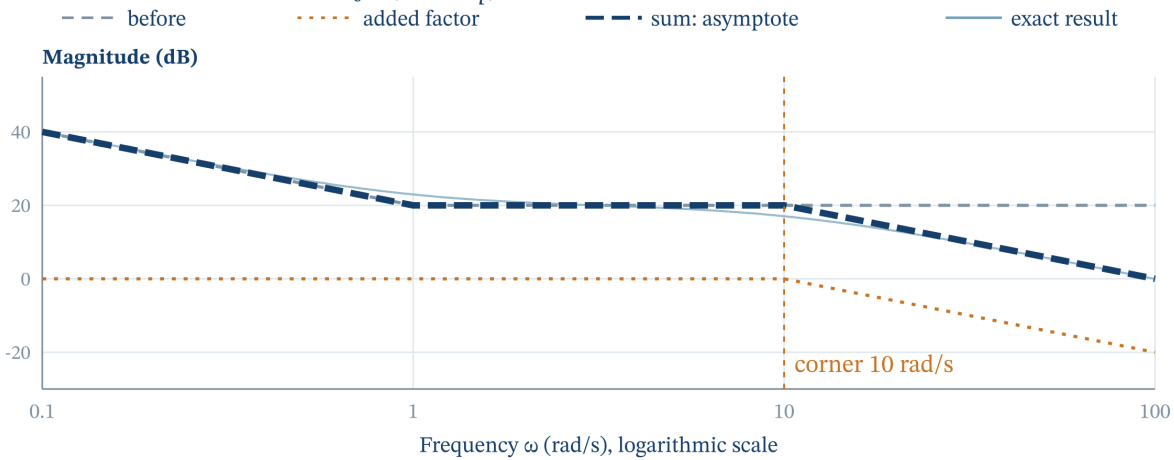
ω	0.1	1	10	100
Before	40	20	0	−20
Zero	0	0	+20	+40
Sum	**40**	**20**	**20**	**20**

Zero adds +20 dB/dec after 1 rad/s.

Result: −20 dB/dec below 1, then flat.

Add a pole at 10 rad/s

$$G_3(s) = \frac{10\,\omega_0\,\omega_p\,(s+\omega_z)}{\omega_z\,s\,(s+\omega_p)} \qquad \omega_z = 1,\ \omega_p = 10 \text{ rad/s}$$



ADD MAGNITUDES IN dB

ω	0.1	1	10	100
Before	40	20	20	20
Pole	0	0	0	−20
Sum	**40**	**20**	**20**	**0**

Pole adds −20 dB/dec after 10 rad/s.

Result: −20 below 1; flat from 1 to 10; −20 above 10 dB/dec.

Two-pole Bode calculation

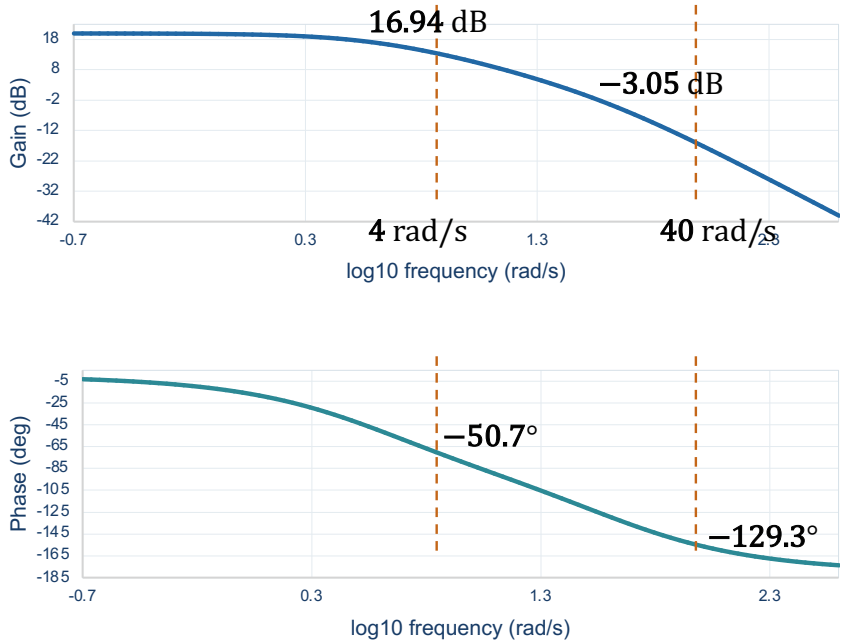
EXAMPLE

Model read from the sweep

20 dB at DC: $K = 10$. Corners 4 and 40 rad/s: two real poles; no integrator.

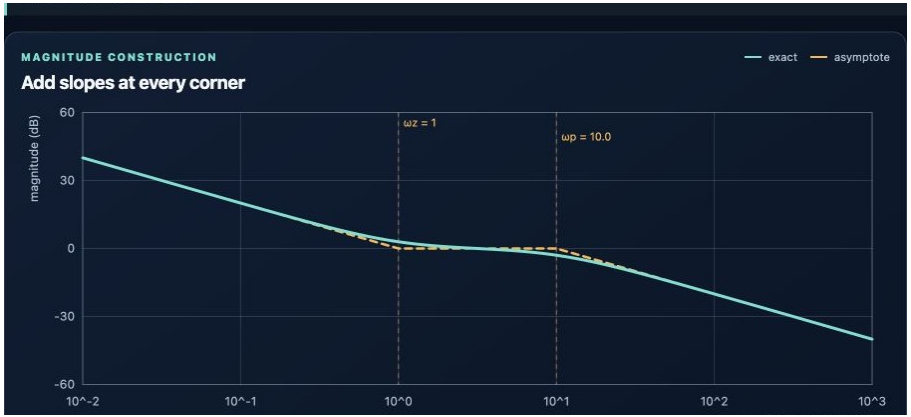
Exact checks: at 4 rad/s the magnitude is 16.94 dB and phase −50.7°. At 40 rad/s the magnitude is −3.05 dB and phase −129.3°.

The phase checks the factor interpretation. **A magnitude-only fit could miss a delay** or a non-minimum-phase zero.



Live demo: build and check the Bode sketch

OPEN LIVE DEMO



Course home: Demo 5.

SET UP Lead network with Integrator ($n = 1$). $K = 1$, $\omega z = 1$, $\omega p = 10$ rad/s.

1 Hide exact response. Predict the slopes in the three frequency regions.

2 Reveal exact response. Inspect the corners.

3 Set $K = 2$. Explain the 6.02 dB shift. MATLAB draws the same two curves.

MATLAB alternative (base MATLAB)

```
K=1; w=logspace(-2,3,500);
H=K*(1+1i*w)./(1i*w.*(1+1i*w/10));
A=20*log10(K./w.*max(1,w) ...
  ./max(1,w/10));
figure; semilogx(w,20*log10(abs(H)), ...
  w,A,'--'); grid on
xlabel('rad/s'); ylabel('dB')
legend('Exact','Asymptote')
```

CHECK $G(s) = \frac{1+s}{s(1+s/10)}$. Slopes: −20, 0, −20 dB/dec. Doubling K adds 6.02 dB without changing phase or corner frequencies.

Reading a Bode plot

CORE

Read this	From	It gives you
Low-frequency level	the flat left-hand end	the DC gain K
Low-frequency slope	$0, -20$ or -40 dB/dec	the number of integrators
Each slope change	the corner frequency	a pole or a zero, and where
A peak	height above the low-frequency level	a lightly damped pair, small ζ
The -3 dB point	relative to the DC level	the bandwidth

PART 2

Resonance and model validation

01 Second-order frequency response

02 Mechanical parameter identification

03 First- and second-order bandwidth

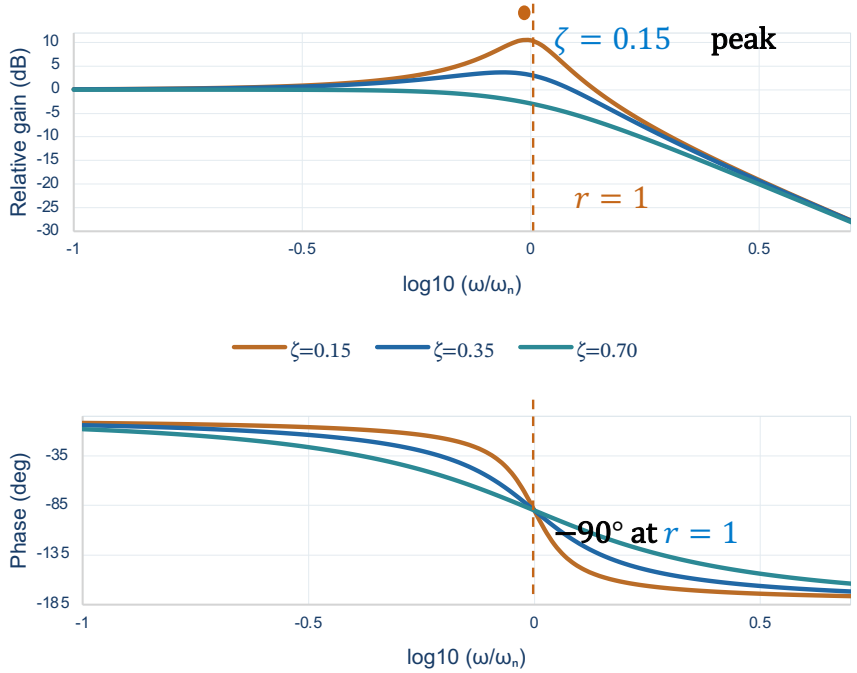
Second-order frequency response

Normalised second-order plant

Let $r = \frac{\omega}{\omega_n}$. The denominator becomes $(1 - r^2) + j2\zeta r$. The phase crosses −90° at $r = 1$, while the magnitude may peak below $r = 1$.

A finite resonant peak exists only for ζ<1/√2. At high frequency the magnitude falls at −40 dB per decade and phase tends to −180°.

Damping $\zeta = 0.15$, 0.35, 0.70; same ω_n and DC gain.



Resonance

Normalised frequency = forcing frequency / natural frequency. Divide magnitude by the positive DC gain.

$$M(r) = \frac{|G(j\omega)|}{K} = \frac{1}{\sqrt{(1-r^2)^2 + 4\zeta^2 r^2}}$$

Maximise magnitude by minimising its squared denominator:

$$D(r) = (1-r^2)^2 + 4\zeta^2 r^2$$

$$\frac{dD}{dr} = 4r(r^2 + 2\zeta^2 - 1) = 0$$

$$r_r = \sqrt{1-2\zeta^2}, \quad \omega_r = \omega_n\sqrt{1-2\zeta^2}$$

Finite resonant peak: positive damping below 0.707; otherwise the maximum is at zero frequency.

$$D(r_r) = 4\zeta^2(1-\zeta^2), \quad M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

EXAMPLE

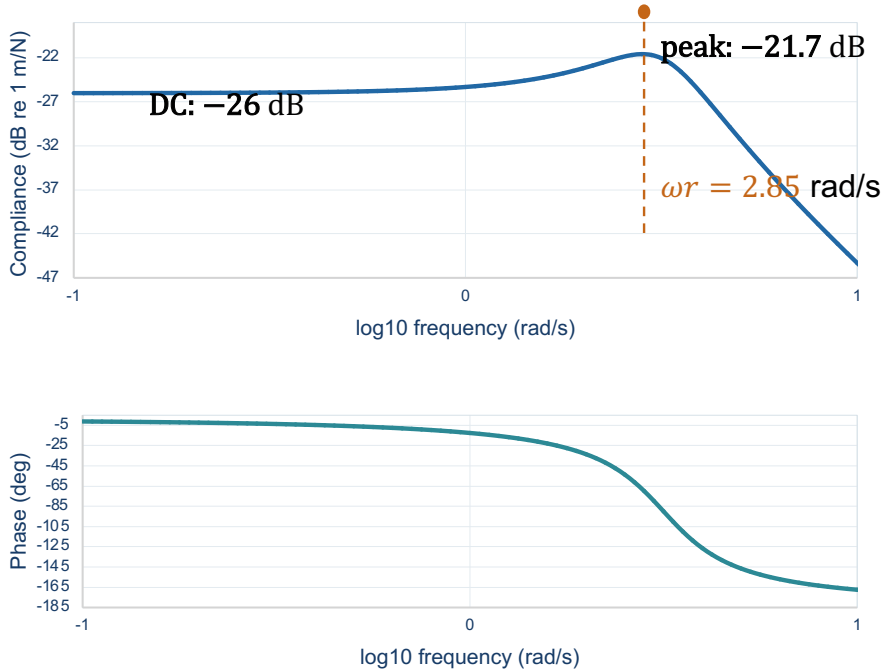
Identify a mechanical plant from a sweep

Source: original MECH321 Lecture 11 (Reference). Readings are marked on the plot.

Measured evidence

Force input; measured displacement. The plotted ideal second-order model reconstructs rounded reference readings.

Peak rise: 4.3 dB above DC. Convert to relative magnification.



Reconstruct a second-order model from Bode data

EXAMPLE

1 Low-frequency compliance

$20\log_{10}(1/k) = -26$ dB gives $1/k \approx 0.0501$ m/N, hence $k \approx 20$ N/m.

2 Resonant peak: smaller root (damping < 0.707)

The 4.3 dB rise means $\mathrm{Mr} = 10^{(4.3/20)} \approx 1.64$. $\zeta \approx 0.32$.

$$M_r^2 = \frac{1}{4\zeta^2(1-\zeta^2)}$$

$$q = \zeta^2, \quad q^2 - q + \frac{1}{4M_r^2} = 0$$

$$\zeta = \sqrt{\frac{1-\sqrt{1-M_r^{-2}}}{2}} = 0.322$$

3 Frequency scale and inertia

From $\omega_r = 2.85$ rad/s and $\omega_r = \omega_n\sqrt{1-2\zeta^2}$, obtain $\omega_n \approx 3.20$ rad/s. Then $m = \frac{k}{\omega_n^2} \approx 2.0$ kg (rounded).

4 Damping and reconstructed model

$b = 2\zeta\omega_n m \approx 4.0$ N·s/m. Recalculate the full magnitude and phase curves with these parameters; **agreement only at the peak is insufficient**.

Final fit uses m≈1.95 kg:

$$G(s) \approx \frac{1}{1.95s^2 + 4.0s + 20}$$

Check the reconstructed model

Second-order hypothesis

A dominant complex pair sets the peak. Keep the measured DC gain.

The sweep gives inertia 1.95 kg, damping 4.0 N·s/m and stiffness 20 N/m.

Bode verification

Overlay model and measured magnitude and phase over the measured frequency band.

Check DC level, peak height and frequency, phase crossing and high-frequency slope.

Independent step test

The fitted model predicts about 35% overshoot, 1.04 s to first peak and 3.9 s to settle.

Compare with a measured step trace if one is available.

Bandwidth and speed

For a stable first-order low-pass plant with positive DC gain, bandwidth is the frequency where amplitude falls to 70.7% of the DC level (−3.01 dB).

$$G(s) = \frac{K}{\tau s + 1}, \quad \frac{|G(j\omega)|}{K} = \frac{1}{\sqrt{1 + (\omega\tau)^2}}$$

Set the normalised amplitude to the bandwidth threshold:

$$\frac{1}{\sqrt{1 + (\omega_b\tau)^2}} = \frac{1}{\sqrt{2}} \qquad 1 + (\omega_b\tau)^2 = 2, \quad \omega_b = \frac{1}{\tau}$$

For the same first-order step, the 10–90% rise time gives:

$$t_r = \tau \ln 9\,, \quad \omega_b t_r = \ln 9 \approx 2.197$$

Halving the time constant doubles bandwidth and halves rise time. This exact link applies only to standard first-order models.

Second-order bandwidth derivation

EXAMPLE

For a stable standard second-order low-pass model with positive DC gain and no zeros, bandwidth is the frequency at 70.7% of the DC amplitude.

1 Normalise frequency and divide magnitude by DC gain

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad r = \frac{\omega}{\omega_n}$$

$$\frac{|G(j\omega)|}{K} = \frac{1}{\sqrt{(1 - r^2)^2 + 4\zeta^2 r^2}}$$

2 Set the squared denominator to two and use the positive root

$$\left(1 - r_b^2\right)^2 + 4\zeta^2 r_b^2 = 2, \quad q = r_b^2$$

$$q^2 + (4\zeta^2 - 2)q - 1 = 0 \qquad q = 1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2}$$

$$\omega_b = \omega_n\sqrt{1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

Worked comparison: two second-order plants

EXAMPLE

Question Compare these two unity-gain plants: find bandwidth (rad/s), peak time (s) and overshoot. Which follows faster changes?

$$G_1(s) = \frac{100}{s^2 + 12s + 100}, \quad G_2(s) = \frac{625}{s^2 + 30s + 625}$$

Match the standard form, then substitute the common damping ratio.

$$\omega_{n1} = 10, \quad \omega_{n2} = 25, \quad \zeta_1 = \frac{12}{2(10)} = \zeta_2 = \frac{30}{2(25)} = 0.6$$

$$\frac{\omega_b}{\omega_n} = \sqrt{1 - 2(0.6)^2 + \sqrt{4(0.6)^4 - 4(0.6)^2 + 2}} = 1.1482$$

$$\omega_{b1} = 11.482, \quad \omega_{b2} = 28.706$$

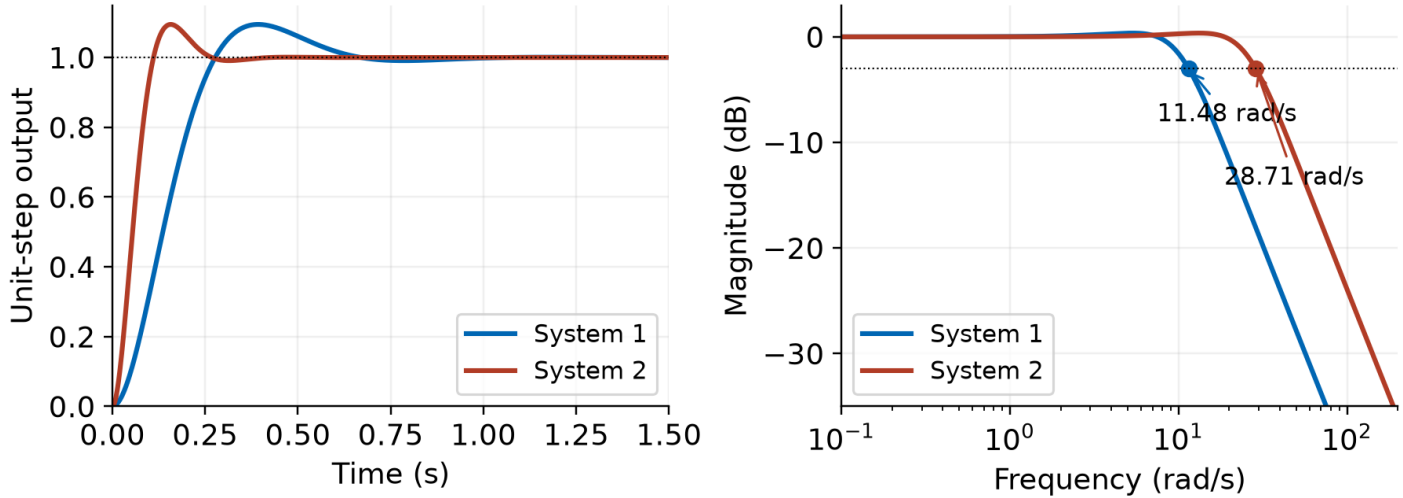
$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} = 0.0948, \quad \frac{\omega_{b2}}{\omega_{b1}} = 2.5$$

$$t_{p1} = \frac{\pi}{10\sqrt{1 - 0.6^2}} = 0.393, \quad t_{p2} = \frac{\pi}{25\sqrt{1 - 0.6^2}} = 0.157$$

Bandwidth and step-response comparison

EXAMPLE

Same final value and 9.48% overshoot. Read bandwidth and first-peak time.



At fixed damping, natural frequency ×2.5 gives bandwidth ×2.5 and peak time ÷2.5. Recheck if damping or numerator changes.

Predict, then simulate

EXAMPLE

Compare a two-pole plant with its one-pole approximation

PREDICTION

For $G = \frac{10}{\left(1+\frac{s}{4}\right)\left(1+\frac{s}{40}\right)}$

Low-frequency level 20 dB

First bend down at 4, second at 40 rad/s

High-frequency slope -40 dB/dec

Phase from 0° to -180°, through -90° near 13 rad/s

SIMULATION

Full plant and dominant-pole model

```
G = tf(10, conv([1/4 1],[1/40 1]));
G1 = tf(10, [1/4 1]);
bode(G,G1); grid on
step(G,G1); grid on
```

When one real pole is enough

Below the next corner: one dominant Bode bend, −20 dB/dec and phase toward −90° support a first-order approximation. A monotonic step alone is insufficient.


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